Rossmoyne Senior High School

Semester One Examination, 2018

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section Two: Calculator-assumed

Name _______________________________________

Teacher's Name _______________________________________

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Materials required/recommended for this section

To be provided by the supervisor
This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate
Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators approved for use in this examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	8	8	50	52	35
Section Two: Calculator-assumed	14	14	100	98	65
				Total	100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer booklet.

3. You must be careful to confine your response to the specific question asked and to follow any instructions that are specified to a particular question.

4. Supplementary pages for the use of planning/continuing your answer to a question have been provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.

5. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.

6. It is recommended that you do not use pencil, except in diagrams.

7. The Formula sheet is not to be handed in with your Question/Answer booklet.

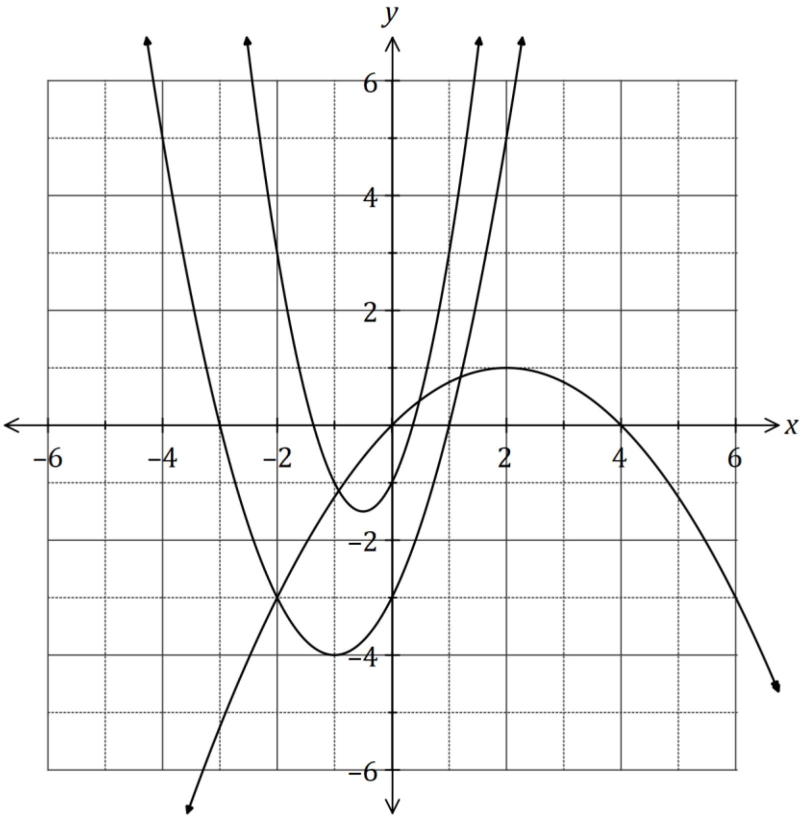
Section Two: Calculator-assumed **65% (98 Marks)**

This section has **fourteen (14)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 9 **(3 marks)**

The graphs of $y = 2x^2 + 2x + c$, $y = a(x-2)^2 + 1$ and $y = (x+b)(x+3)$ are shown below.



Determine the values of the constants a, b and c.

Question 10 **(7 marks)**

An online store is offering new customers the opportunity to select 5 different products for just \$5 each. They can select from a range of 8 different books, 12 different magazines and 16 different games.

Determine how many different selections can be made if:

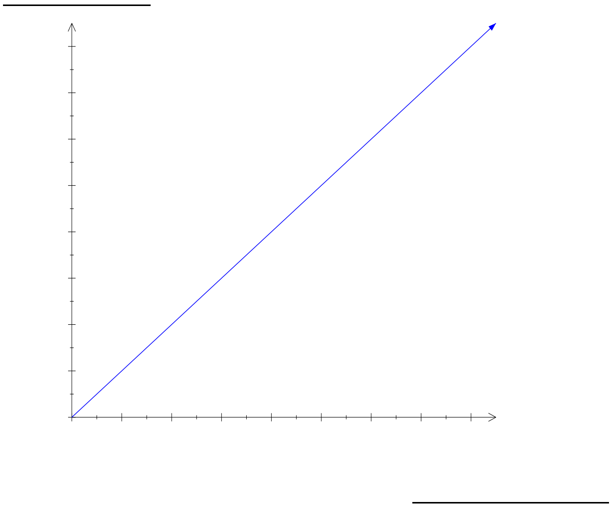
(a) there are no restrictions. (2 marks)

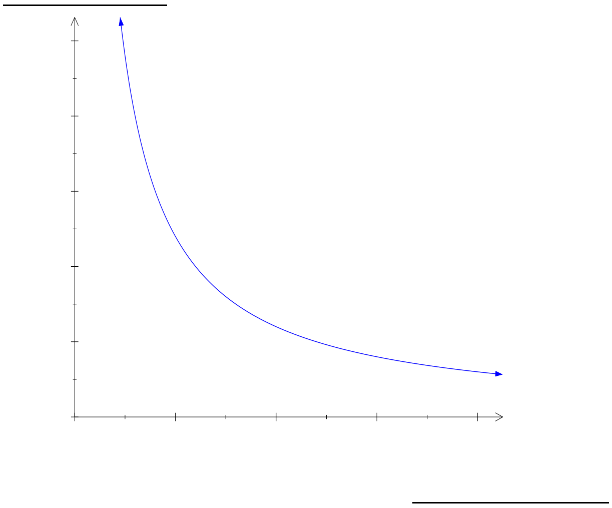
(b) only games are included. (2 marks)

(c) at least 3 games are included. (3 marks)

Question 11 **(6 marks)**

Trevor and Steven have decided to landscape their backyard including laying 80m^2 of artificial lawn at a cost of $4000. They know that the two of them could complete the landscaping in 24 hours and that they have some friends who could help them if required.





(a) One of the graphs shown depicts the total cost of artificial lawn for different areas and the other shows the total time taken for different numbers of people working on the landscaping.

Add the words **COST**, **AREA**, **HOURS** and **PEOPLE** to the axes above. (2 marks)

(b) Trevor and Steve decide to lay only 50m^2 of lawn. How much will this cost? (2 marks)

(c) Write an equation that relates the number of hours taken for the number of people landscaping. Hence, determine how long it would take six people to complete the landscaping. (2 marks)

Question 12 **(8 marks)**

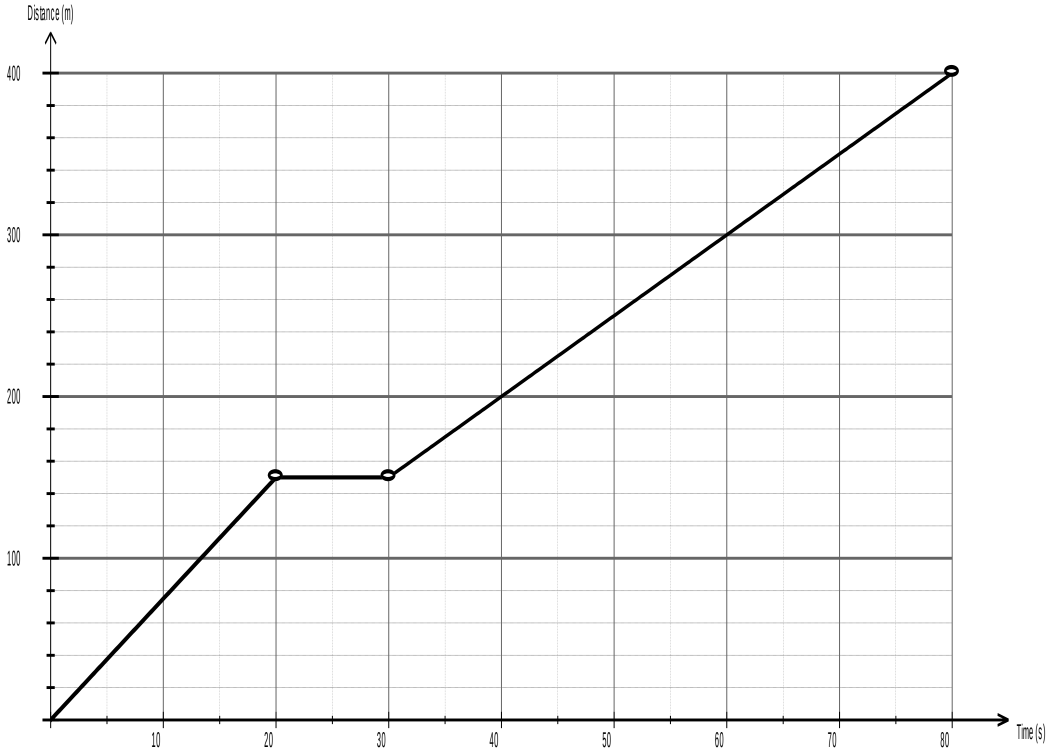
(a) Determine the equation of the axis of symmetry for the graph of $y = 4x^2 - 20x - 51$.
(2 marks)

(b) The graph of $y = ax^2 + bx + 27$ passes through the points $(3, 6)$ and $(-4, -57)$.
Determine the values of the constants a and b. (3 marks)

(c) Determine the value(s) of m for which the graph of $y = x^2 - 4x + m$
has no real roots.
(3 marks)

Question 13 **(6 marks)**

In the television special 'The World's Fittest Netball Player', Laura Geitz and Natalie von Bertouch take part in a 400m challenge. Laura's effort at the 400m course is shown below.



(a) Determine the three different speeds in m/s travelled by Laura through the 400m. (2 marks)

(b) Natalie ran at 21.6km/h for the entire 400m. Determine her speed in m/s and draw her journey on the axes above.

Hence or otherwise determine when and where Natalie overtakes Laura. (4 marks)

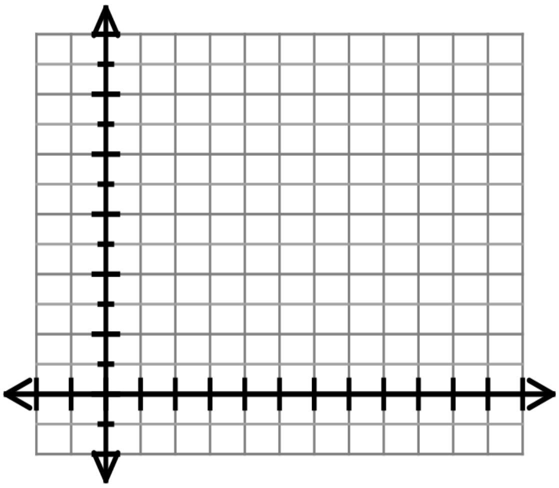
Question 14 **(7 marks)**

The current A (amperes), varies inversely to the resistance R (ohms) in an electric circuit. When the resistance is 12 ohms, the current is 0.5 amperes.

(a) State the constant of variation. (1 mark)

(b) Calculate the resistance if the current is 5 amperes. (1 mark)

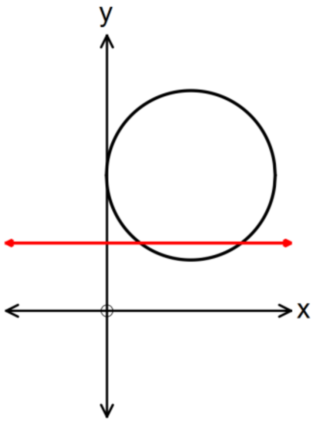
(c) By first clearly plotting 2 points, draw the graph of this relationship on the axes below. Label the axes. (3 marks)



(d) Determine the effect on R if A is increased by 35%. (2 marks)

Question 15 **(8 marks)**

The circle with centre A(5, 8) touches the y – axis as shown below.



(a) By first expressing the equation of the circle in the form $(x-h)^2+(y-k)^2=r^2$ with centre (h , k) and radius r, show clearly that the equation of the circle is $x^2 + y^2 - 10x - 16y + 64 = 0$. (3 marks)

(b) The line $y = 4$ intersects the circle at points M and N.

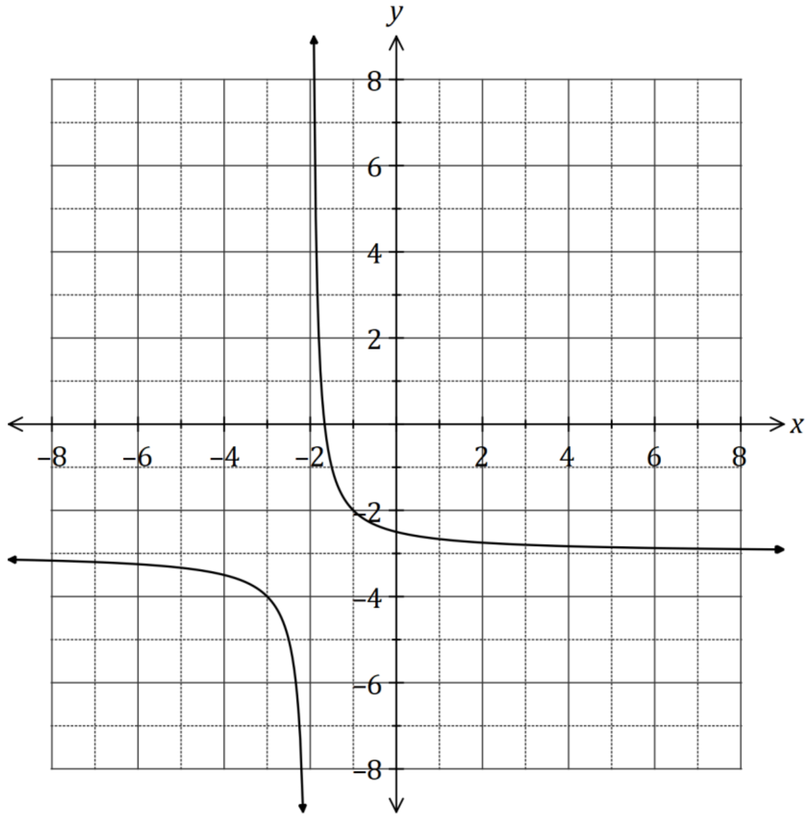
(i) Determine the coordinates of points M and N. (3 marks)

(ii) Determine the length of the chord MN. (2 marks)

Question 16 **(8 marks)**

Let $f(x) = \dfrac{4}{3-x}$ and $g(x) = \dfrac{1}{x+p} + q$, where p and q are constants.

The graph of $y = g(x)$ is shown below.



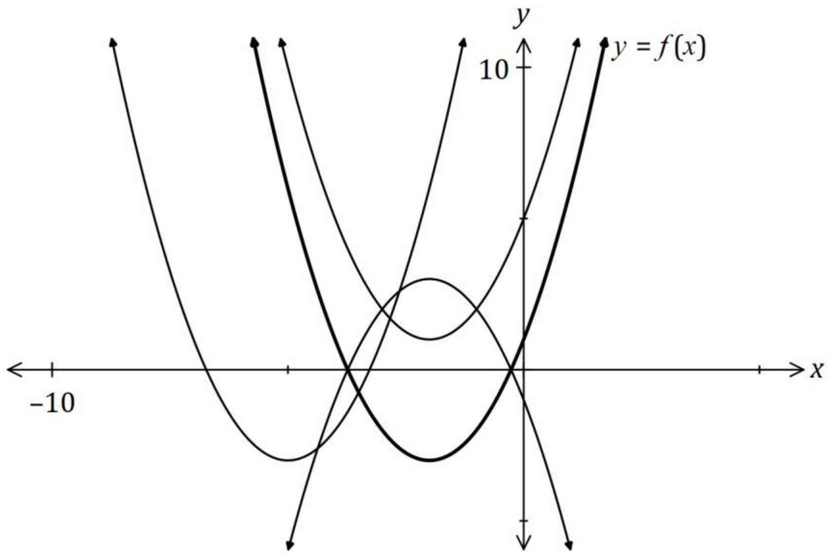
(a) Sketch the graph of $y = f(x)$ on the axes above. Clearly label the asymptotes and y-intercept(s). (3 marks)

(b) Determine the values of p and q. (2 marks)

(c) Solve the equation $f(x) = g(x)$, giving your solution(s) to one decimal place. (3 marks)

Question 17 **(6 marks)**

(a) The graph of $y = f(x)$ is shown in bold below. The graphs of $y = -f(x)$, $y = f(x+p)$ and $y = f(x) + q$ are also shown, where p and q are constants.



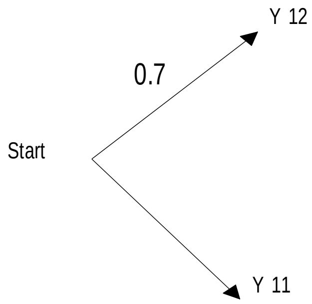
Clearly label the remaining graphs with $y = -f(x), y = f(x+p)$ or $y = f(x) + q$.
(3 marks)

(b) The one-to-one relation $y = 7 - 3x$ has domain and range given by $\{x: x = -2, 3, a\}$ and $\{y: y = -8, -2, b\}$ respectively. Determine the values of constants a and b. (3 marks)

Question 18 **(9 marks)**

In a group of students, it is known that 70% are in Year 12 and the rest in Year 11. 40% of those in Year 12 and 80% of those in Year 11 study maths, while the remainder do not.

(a) Complete the probability tree diagram to represent the above information.



(2 marks)

Determine the probability that a randomly chosen student from the group

(i) is in Year 11 and studies maths. (1 mark)

(ii) studies maths. (2 marks)

(iii) is in Year 11, given that they study maths. (2 marks)

(iv) is in Year 12 or studies maths. (2 marks)

Question 19 **(8 marks)**

For two events, X and Y, it is known that $P(X) = 0.48$ and $P(Y) = 0.35$. Determine the following probabilities, **using any additional information only within that part of the question**.

(a) $P(\bar{Y})$. (1 mark)

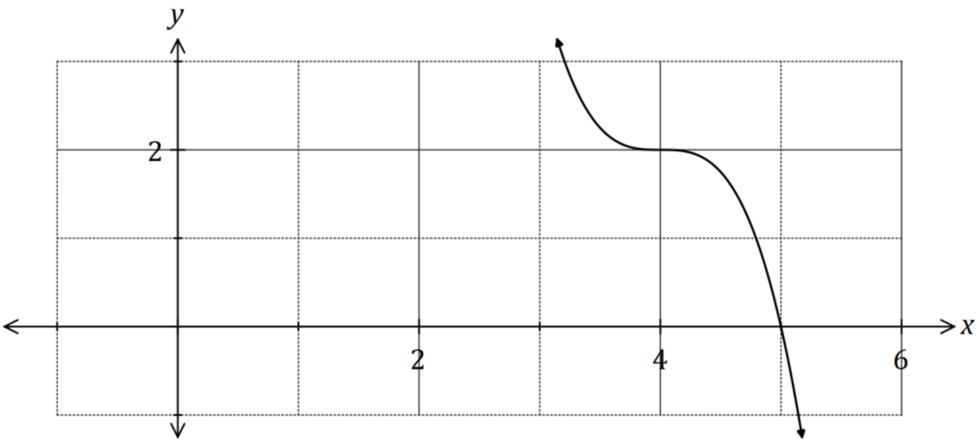
(b) $P(Y \cap \bar{X})$ when $P(X \cap \bar{Y}) = 0.29$. (2 marks)

(c) $P(X|Y)$ when $P(Y|X) = 0.7$. (3 marks)

(d) $P(X \cup Y)$ when X and Y are mutually exclusive. (2 marks)

Question 20 **(6 marks)**

(a) Part of the graph of $y = f(x)$ is shown below, where $f(x) = -2(x-b)^3 + c$, and b and c are constants.



(i) State the degree of $f(x)$. (1 mark)

(ii) Determine the value of b. (1 mark)

(iii) Determine $f(0)$. (2 marks)

(b) Another function is given by $g(x) = -f(x+8)$.

Describe the transformation that maps the graph $y = f(x)$ onto $y = g(x)$. (2 marks)

Question 21 **(9 marks)**

Injury events in a state were grouped as follows: (D) injury events resulting in death; (H) injury events requiring hospitalisation; and (E) injury events requiring treatment at an emergency department. Injury events were further grouped according to whether prior use of alcohol was a contributing factor. The number of injury events for one month is summarised in the table below.

	D	H	E	**Total**
Alcohol	25	581	5 605	**6 211**
No Alcohol	115	4 422	11 912	**16 449**
Total	**140**	**5 003**	**17 517**	**22 660**

(a) Calculate the percentage of all injury events that involved prior use of alcohol. (2 marks)

(b) If one injury event was selected at random from the data, determine the probability that

(i) it required hospitalisation. (1 mark)

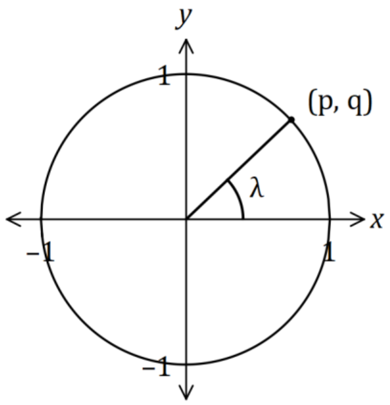
(ii) it did not require treatment at an emergency department given that prior use of alcohol was a contributing factor. (2 marks)

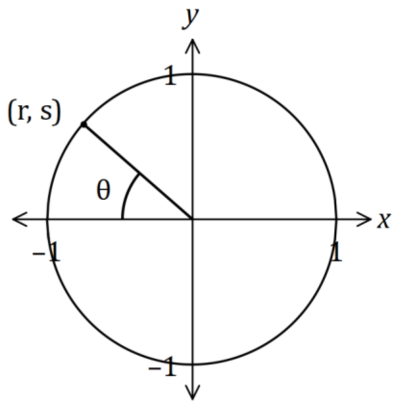
(iii) prior use of alcohol was a contributing factor given that it required hospitalisation. (2 marks)

(c) Use your answers above to show that alcohol and hospitalisation are not independent. (2 marks)

Question 22 **(7 marks)**

Consider the points with coordinates (p, q) and (r, s) that lie in the first and second quadrants respectively of the unit circles shown below, where λ and θ are acute angles.





Determine the following in terms of p, q, r and s, simplifying your answers where possible.

(a) $\sin\theta$. (1 mark)

(b) $\tan\theta$. (2 marks)

(c) $\cos(\pi + \lambda)$. (2 marks)

(d) $\sin(90° - \lambda)$. (2 marks)

End of questions

Supplementary page

Question number: _________

Supplementary page

Question number: _________